

LAB TEST
CSE3023/CSA3023/CSM3023
2025/2026-2

PART A: CASE STUDY - DRIVESMART INTEGRATED ACADEMY

DriveSmart is an integrated, web-based driving school management platform. Unlike traditional schools, DriveSmart operates a centralized network where registered students can book and attend their driving training sessions at *any* branch nationwide.

You are hired as a Junior Web Developer to build a prototype for the "**Session Booking**" and "**Centralized Schedule**" modules, unified by a central **Landing Page**. The system must utilize a strict Model-View-Controller (MVC) architecture to ensure the separation of business logic (Java/Servlet) from the presentation view (HTML/JSP).

Scenario Requirements:

1. **Unified Landing Page:** A central menu providing navigation to all modules, utilizing shared static headers and dynamic footers.
2. **Booking Module:** Students or admins must be able to register a new driving lesson via a web form, selecting their preferred branch location.
3. **Schedule Module:** Instructors and admins must be able to view a real-time, centralized table of all booked training sessions retrieved from the database across all branches.

PART B: STEP-BY-STEP INSTRUCTIONS (100 Marks)

All tasks **must be** implemented within a single project.

Task 1: Database & Model Preparation (15 Marks)

1. **Database Setup (5 Marks):** Create a MySQL database named `drivesmart_db`. Create a table named `Training_Sessions` with the following columns:
 - a. `session_id` (Primary Key, Auto Increment)
 - b. `student_name` (String/Varchar)
 - c. `branch_location` (String/Varchar - e.g., "Kuala Lumpur", "Penang", "Johor")
 - d. `lesson_type` (String/Varchar - e.g., "Manual Car", "Automatic Car", "Motorcycle")
 - e. `status` (String/Varchar - e.g., "Booked", "Completed")
2. **JavaBean/Model (10 Marks):** Create a JavaBean class named `SessionBean.java`.
 - a. Declare all the corresponding private variables.
 - b. Generate an empty constructor and a parameterized constructor.
 - c. Generate standard getter and setter methods for all variables.

Task 2: Database Connection & DAO (15 Marks)

1. Create a Data Access Object (DAO) Java class named `SessionDAO.java`.
2. **Connection Method (5 Marks):** Write a method to establish a JDBC connection to the `drivesmart_db` database.
3. **Insert Method (5 Marks):** Write a method `bookSession(SessionBean session)` that executes an SQL INSERT statement to save a new training booking into the database.
4. **Retrieve Method (5 Marks):** Write a method `getAllSessions()` that executes an SQL `SELECT * FROM Training_Sessions ORDER BY branch_location ASC` statement and returns a `List<SessionBean>`.

Task 3: Controller (Servlets) Development (20 Marks)

1. **Booking Controller (10 Marks):** Create a Servlet named `BookSessionServlet`.
 - o Use the `doPost()` method to retrieve parameters (`student_name`, `branch_location`, `lesson_type`) from the HTML form.
 - o Instantiate `SessionBean`, set the retrieved values (default status to "Booked"), and pass the object to the `SessionDAO` to save it.
 - o Upon success, use `response.sendRedirect()` to redirect the user to the Schedule Servlet.

2. **Schedule Controller (10 Marks):** Create a Servlet named ScheduleServlet.
 - Use the doGet() method to call getAllSessions() from your DAO.
 - Set the retrieved list as a request attribute: request.setAttribute("sessionList", list);.
 - Use the <jsp:forward> logic via RequestDispatcher to forward the HTTP request to the JSP schedule page.

Task 4: View (JSP & UI) Development (20 Marks)

1. **Book Session View (10 Marks):** Create book_session.jsp.
 - Design an HTML <form> that uses method="POST" and action="BookSessionServlet".
 - Include input fields for Student Name, Branch Location, and Lesson Type.
2. **Centralized Schedule View (10 Marks):** Create schedule.jsp.
 - Design an HTML <table> to display the training data.
 - Use JSP Scriptlets (<% for(...) { %>) or JSTL (<c:forEach>) to iterate through the sessionList attribute and print the details of each training session inside the table rows.

Task 5: Centralized Landing Page Integration (20 Marks)

1. **Shared UI Components (5 Marks):** Create a static header.html (containing the DriveSmart Academy logo/title) and a dynamic footer.jsp (displaying the current server date/time using a JSP Expression: <%= new java.util.Date() %>).
2. **Landing Page Navigation (15 Marks):** Create a main index.jsp file. You must strictly apply the following rules:
 - Use the **JSP Include Directive** (<%@ include file="..." %>) to merge the header.html and footer.jsp into the top and bottom of the page.
 - Use standard HTML anchor tags (<a>) to create a central navigation menu pointing to the two functional modules: the Booking Form (book_session.jsp) and the Schedule Dashboard (ScheduleServlet).

Task 6: MVC Architecture & Submission Best Practices (10 Marks)

- **Separation of Concerns:** Ensure absolutely NO raw SQL queries or direct JDBC connections are written inside the .jsp files.
- **Folder Structure:** Ensure Java classes (Beans, DAO, Servlets) are in proper packages and JSP files are placed in the correct WebContent or Web Pages directory.
- **Submission Format:** The code must be successfully hosted on GitHub and submitted properly (details in Part C).

PART C: SUBMISSION METHOD

To ensure an easier and more efficient project evaluation, all completed lab tests must be submitted using the **GitHub** method.

Steps for Submission:

1. **Host on GitHub:** Push your complete web-based system source code (including your Servlets, JavaBeans, DAOs, JSPs, and HTML files) to a repository on GitHub.
2. **Prepare Document:** Copy the URL link of your GitHub project repository. Paste this URL into a new Word document along with your Name and Matric Number.
3. **Upload to ePembelajaran:** Submit this Word document to the designated **ePembelajaran** submission link (by respective group) before the 2 hours and 30 minutes time limit expires.
4. **However,** once the exam period is over, the remaining assessment on the project output **consequently depends directly** on their respective lecturers.